



[83]

SARDAR PATEL UNIVERSITY

External Examination

M. Sc. (IT) (Integrated) 1st Semester (CBCS) (NC)

PS01CIIT03: Problem Solving Methodology and Programming in 'C'

11th February, Friday, 2022

Time: 12.00pm to 2.00pm

Total Marks: 70

Q-1 Select an appropriate option.

[10]

1. _____ is a step by step approach to solve any problem.
(a) Process (b) Programming Language (c) Algorithm (d) Compiler
2. _____ is a pictorial representation of an algorithm.
(a) Data Diagram (b) Flowchart (c) Pie Chart (d) Program
3. The process of walking through a program's logic on paper before you actually write the program is called _____.
(a) Desk checking (b) Flowcharting (c) Pseudocoding (d) Testing
4. The value can be changed during program execution is known as _____.
(a) Variable (b) constant (c) Operator (d) None of these
5. The combination of ? and : is known as _____ operator.
(a) Ternary (b) Arithmetic (c) dot (d) Relational
6. _____ format specifier used for integer data type.
(a) %d (b) %f (c) %c (d) %e
7. _____ loop execute at least once.
(a) While loop (b) do...while (c) if (d) for
8. For declaring one-dimensional array _____ subscript is used.
(a) One (b) Two (c) Three (d) All of these
9. A group of character is known as _____.
(a) String (b) Array (c) Function (d) All of these
10. _____ operation is performed on string.
(a) Reading (b) Copying (c) Writing (d) All of these

Q-2 Fill in the blanks. / State True or False

[08]

1. C language is _____ level language.
2. _____ is easily re-locatable language.
3. _____ statement terminates the execution of loop.
4. _____ format specifier used for floating data type.
5. For declaring two-dimensional array two subscript is used. (True / False)

[1]

[P.T.O.]

6. '\0' is called Null Character. (True / False)
7. printf() function is used to reading the string. (True / False)
8. Ampersand (&) is not required when read the string value. (True / False)

Q-3 Answer the following questions. (Attempt any **TEN**)

[20]

1. What is an algorithm? List characteristics of an algorithm.
2. Write advantages and disadvantages of an algorithm.
3. Write advantages and disadvantages of flow chart.
4. Draw the basic structure of C program.
5. Write Rules of variable name.
6. Explain printf() statement with example.
7. Explain continue statement in C.
8. Explain the isalpha() function.
9. Explain the sqrt() function.
10. What is string? Explain.
11. Explain gets() function with syntax and example.
12. Write difference between user define function and library function.

Q-4 Answer the following questions. (Attempt any **FOUR**)

[32]

1. What is Flowchart? Explain rules to draw flowchart. Also explain symbols used to draw flowchart.
2. Write an algorithm for following.
 - (i) To find out minimum from N numbers.
 - (ii) To check whether inputted number is prime number or not.
3. What is operator? Explain different operators with example.
4. Explain if..else & nested if statement with syntax and example.
5. Explain while and for loop with syntax and example.
6. Define array? Also explain declaration of 1D and 2D array with syntax and example.
7. Define function? Explain type of function use in c with example.
8. Explain functions: (i) strlen() (ii) strcat() (iii) strcmp (iv) strcpy

————— ✕ —————